Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords:

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Krishna Mishra
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**Lab: Generative AI**
Semester	**7**	**Course Code**	**23ADS1706**
Roll No	**41**	**Name of Student**	**Krishna Mishra**

Practical Number	**6**
Course Outcome	CO1: Explain the fundamentals of Artificial Intelligence, Neural Networks, Natural Language Processing, and Transfer Learning used in Generative AI applications. CO2: Apply Generative AI models such as GANs, VAEs, Transformers, BERT, GPT, and T5to solve real-world AI and NLP problems. CO3: Analyze and optimize prompts using prompt engineering techniques for improving the performance of Large Language Models in different application domains. CO4: Design and develop Generative AI solutions by integrating Foundation Models, Retrieval-Augmented Generation (RAG), tokenization, fine-tuning, and chunking techniques for domain-specific applications.
Aim	To generate concise abstractive summaries of lengthy news articles using a transformer-based model (DistilBART).
Problem Definition	A news portal needs automatic summaries of lengthy news articles. Generate concise summaries using transformer-based models.
Theory (100 words)	Text summarization produces a short version of a long document that keeps its main ideas. Extractive methods copy important sentences, whereas abstractive methods generate new sentences in their own words. The Transformer architecture uses an encoder–decoder structure with self-attention, which lets every word attend to all other words of the article. BART is a transformer encoder–decoder pre-trained to reconstruct text, and DistilBART is its smaller and faster version fine-tuned on the CNN/DailyMail news dataset. The article is tokenized, the model generates the summary using beam search, and the tokens are decoded back to text.
Procedure	Algorithm: • Start.

and
Execution
(100 Words)

- Load the pre-trained DistilBART summarization model and its tokenizer.
- Store the news article text.
- Tokenize the article and give it to the encoder.
- Generate the summary using the decoder with beam search and minimum/maximum length limits.
- Decode the generated tokens into text.
- Display the original and summary lengths and the summary.
- Stop.

Code:

```
import torch
from transformers import AutoTokenizer,
AutoModelForSeq2SeqLM

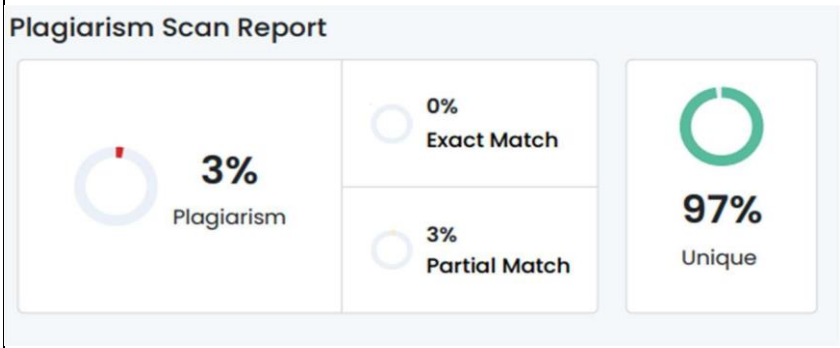
model_name = "sshleifer/distilbart-cnn-12-6"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model =
AutoModelForSeq2SeqLM.from_pretrained(model_name)

article = """
The city council has approved a new metro rail project that will
connect the eastern suburbs with the central business district.
The project is expected to cost nearly 8,000 crore rupees and will
include 18 stations along a 24 kilometre route. Officials said the
metro will reduce travel time from almost 90 minutes to about
35 minutes and will help cut traffic congestion on the main
roads.
Construction is scheduled to begin early next year and the first
phase is expected to open within four years. Environmental
groups
welcomed the plan but asked the council to protect existing trees
and green areas during construction. The council has promised to
plant two trees for every one that is removed and to publish
quarterly progress reports for residents.
"""


def summarize(text, max_len=60, min_len=25):
    inputs = tokenizer(
        text,
        return_tensors="pt",
        truncation=True,
        max_length=1024
    )

    with torch.no_grad():
        ids = model.generate(
            inputs["input_ids"],
            num_beams=4,
            max_length=max_len,
            min_length=min_len,
            early_stopping=True
        )
```

	return tokenizer.decode(ids[0], skip_special_tokens=True) summary = summarize(article) print("Original length:", len(article.split()), "words") print("Summary length:", len(summary.split()), "words") print("\nSummary:") print(summary)
Output Analysis	• Input: A news article of about 120 words about a new metro rail project is given to the model. • Output: The model generates an abstractive summary of about 25 to 60 words that keeps the main facts of the article (the approval of the metro project, its route, cost and expected benefits) and leaves out minor details. Record the generated summary from your run here. • Analysis: The summary is much shorter than the original article, so the reader can understand the news quickly. The length can be controlled using the min_length and max_length parameters.
Link of student Github profile where lab assignment has been uploaded. Plag Report (Similarity index < 12%)	Github.com/krish-na45/GenAI-Practicals 
Conclusion	The experiment generated concise summaries of a news article using the transformer-based DistilBART model. The self-attention mechanism helped the model identify the important information and write a short abstractive summary. Such a system can be used by news portals to provide quick summaries of long articles.

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